1. State the role of digital circuit technology in modern communication systems.
2. Compare analog and digital circuits, highlighting their advantages, disadvantages, and practical applications.
3. Analyze the impact of logic minimization on circuit cost, speed, and power consumption.
4. Is minimizing the number of logic gates always the best optimization strategy? Explain with examples.
5. Design a logic circuit with inputs P, Q, R so that output is high whenever P is 0 or whenever Q=R=1. Use NAND gates only.
6. Design a logic circuit that has four inputs (A, B, C, and D) and an output (Y), such that, the output is a 1 whenever the decimal equivalent of $ABCD)_2$ is divisible by either 3 or 5.
7. Design a logic circuit that accepts a 3-bit number and generates an output binary number equal to the square of the input number.
8. Design a logic circuit that accepts a 4-bit number and generates the 2's complement of the input number.
9. Compare combinational and sequential logic circuits with respect to design, operation, and applications.
10. Design and explain the circuit to add the following bits 1011 and 1100 using block- diagram.
11. A standard 8-to-3 encoder receives two active inputs simultaneously. What output will it produce? Why is this a problem?
12. Implement a 1-to-16 demultiplexer using only 2-to-4 decoders with enable inputs and no other logic gates. Clearly label all inputs, pins, and outputs of your circuit.
13. Design a logic circuit that compares two 4-bit numbers, A and B, to check if they are equal. The circuit has one output x, so that x =1 if A = B and x = 0 if $A \neq B$.
14. Explain how a multiplexer can be used as a universal logic circuit.
15. Design a circuit to distribute a serial input to four different outputs using a DEMUX.
16. Give a real-life application where a priority encoder is preferred over a standard encoder.
17. Compare a decoder with a demultiplexer. Under what conditions can a decoder function as a demultiplexer?
18. A logic circuit is defined by the following functions:

$$f_1 = \overline{A}\overline{B} + AB\overline{C}$$
$$f_2 = \overline{A} + B$$
$$f_3 = AB + \overline{A}\overline{B}$$

Draw the circuit with a decoder and external gates.

19. Under what conditions would you choose a D flip-flop instead of a JK flip-flop? Justify your answer.
20. Compare SR, JK, D, and T flip-flops in terms of functionality, complexity, and typical applications.
21. Compare edge-triggered and level-triggered flip-flops. Which is more suitable for high-speed applications, and why?
22. Is a JK flip-flop always a better choice because it is more versatile? Defend your answer.
23. Under what circumstances would you choose a synchronous design over an asynchronous design? Justify your answer.
24. Under what circumstances would an asynchronous counter be a better choice than a synchronous counter?
25. Why are synchronous counters preferred in modern high-speed processors?

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26. Why are synchronous counters generally preferred over asynchronous counters in high-speed digital systems? Discuss the trade-offs involved.
27. Compare ripple counters and synchronous counters in terms of speed, complexity, power consumption, and applications.
28. Which type of counter would you recommend for a high-speed communication system? Justify your answer.
29. Design a counter that counts only even numbers from 0 to 14.
30. Explain a 4-bit asynchronous binary counter with timing diagram. If each flip-flop has a propagation delay of 10ns, determine the total propagation delay time and maximum clock frequency at which the counter can be operated.

31. A certain memory has a capacity of 16Kx32. How many words does it store? What is the number of bits per words? How many different addresses are required by the memory?
32. How many address inputs, data inputs, and data outputs are required for a 16Kx12 memory?

33.

a) How many 32Kx8 RAM chips are needed to provide a memory capacity of 256K bytes?
b) How many lines of the address must be used to access 256K bytes? How many of these lines are connected to the address inputs of all chips?
c) How many lines must be decoded for the chip select inputs? Specify the size of the decoder.

34. Suppose you have 16x4 memory chips available. But you need to create a 64x8 memory unit. i) How many chips are required? ii) Show the external connections necessary to construct the memory unit.
35. Given a 32x8 ROM chip with an enable input. Show the external connections necessary to construct a 128x8 ROM with four chips and a decoder.
36. Analyze the advantages of programmable logic over fixed-function digital circuits.
37. Compare CRC and Hamming code. Under what conditions is CRC preferred, and when is Hamming code more suitable?
38. Why is CRC preferred over a simple parity check?
39. Under what conditions can a parity checker indicate that data is correct even though transmission errors occurred? Justify your answer.
40. Design a sequential circuit with two D flip flops and two inputs, P and Q. If P = 0, the circuit remains in the same state regardless of the value of Q. When P = 1 and Q = 1, the circuit goes through the state transitions from 00 to 01 to 10 back to 00, and repeats. When P = 1 and Q = 0, the circuit goes through the state transitions from 00 to 10 to 01 back to 00, and repeats, The circuit is to be designed by treating the unused state (s) as don't care condition(s).
41. (a) Draw a logic circuit that accepts a 3-bit input number A (A2, A1, A0) and outputs a 2-bit number B (B1, B0) that is equal to the number of 1's that appear in the input A. For example, A=101 contains two 1's, so B=10. Also, A=111 contains three 1's, so B=11. You are allowed to use only one 3-8 decoder, two OR gates, and one 4-2 encoder. Label all inputs, pins and outputs. (b) Explain your solution.

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42. (a) Draw a logic circuit that uses the 4-bit adder shown below and any other basic logic gates that you think. You might need to compute the equation P=3*Q+1. Assume that you have a 3-bit input Q (Q2, Q1, Q0) and a 5-bit output P(P4, P3, P2, P1, P0). Clearly label all inputs and outputs. (b) Explain your solution.

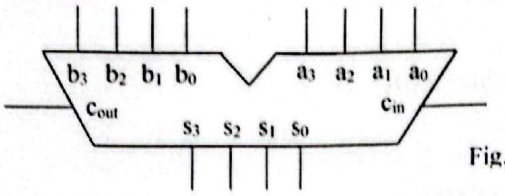


Fig. 4-bit adder

43.

(a) Draw the truth table for the function $f = x'y + x'z + xy'z'$

(b) Implement this function using only 2-to-1 multiplexers and no other logic gates. The inputs to your circuits can be only the signals X, Y, and Z in their non-inverted form.

44.

(a) Draw the truth table for a full-adder.

(b) Implement a full-adder circuit using one 3-to-8 decoder and a minimal number of basic logic gates. Clearly label all inputs, pins, and outputs of your circuit.

45.

Design a circuit that tests if a 4-bit binary number is a prime number in the decimal system. Assume that the input is represented by the signals a, b, c, and d (where a is the most significant bit and d is the least significant bit) and the output is called *f*. Hint: 0 and 1 are not prime.

(a) Draw the truth table for the function *f*(a, b, c, d).

(b) Implement the function *f* using one 8-to-1 multiplexer and a minimal number of extra AND, OR, and NOT gates. Clearly label all inputs, pins, and outputs.

(c) Draw the truth table for the function *f* again on this page. Now, implement *f* using one 4-to-1 multiplexer and a minimal number of extra AND, OR, and NOT gates. Clearly label all inputs, pins, and outputs.

46.

An 8×1 multiplexer has inputs A, B and C connected to the selection inputs S_2, S_1, and S_0 respectively. The data inputs I_0 to I_7 are as follows:

$I_1=I_2=I_7=0$; $I_3=I_5=1$; $I_0=I_4=D$ and $I_6=D'$

(a) Show the multiplexer implementation.

(b) Determine the Boolean function that the multiplexer implements.

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